VirCap: Virtual Camera Exposure Control based on Image Photometric Synthesis for Visual SLAM Application

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***Abstract*— Mobile robots, such as quadrupedal and vehicular robots, are known for their high-speed movement and operation in environments with wide dynamic ranges. This property challenges the existing camera capture methods for visual applications, especially the visual Simultaneous Localization and Mapping (SLAM) task which requires a strong temporal continuity. Due to the limitations imposed by the camera hardware's control frequency and delay, camera exposure control methods cannot rapidly and stably publish high-quality images preventing oversaturation, background noise, and motion blur.**

In this paper, we propose a novel image acquisition framework called VirCap, introducing image bracketing capture patterns to preserve more dynamic range information than a single image capture. By leveraging image photometric synthetic technology, VirCap enables a virtual camera exposure control (exposure time and analog gain), effectively decoupling the camera interaction from the control loop and facilitating more frequent exposure updates than traditional camera control methods. An exposure allocation strategy is also developed to balance motion blur and background noise, allowing VirCap to synthesize images of optimal quality that considers the robot's self-motion. Extensive experiments are conducted to demonstrate the efficiency and resilience of VirCap under extreme operating conditions for different visual SLAM systems.

***Index Terms*— Camera exposure control, Photometric image synthesis, Visual simultaneous localization and mapping, Robotic perception.**

I. Introduction

MOBILE robot industry is experiencing a surge in diverse perception sensors. For the Simultaneous Localization and Mapping (SLAM) task, novel sensors, such as frequency-modulated continuous-wave (FMCW) LiDARs [1] and event cameras [2], [3], have achieved brilliant performance in specific scenarios and working conditions. Frame cameras remain prevalent in visual SLAM applications, benefiting from their rich semantic information and cost-effectiveness. However, compared with other visual tasks such as object detection and semantic segmentation, visual SLAM tasks require better feature information extraction and inter-frame information association, which places higher requirements on camera performance. Despite years of advancement, frame camera perception systems for moving robots continue to encounter challenges and exhibit fragility and sensitivity owing to inherent characteristics:

- The *dynamic range* of natural scenes exceeds 100 dB, which is significantly broader than the approximately 70 dB range that global shutter frame cameras can capture. Information beyond this range is lost due to the oversaturation, including underexposure or overexposure.
- *Scene illuminance* can change drastically when robots move rapidly through areas with varying lighting conditions. Additionally, luminance can change actively with alterations in the light source.
- The high mobility of mobile robots often results in *motion blur*, particularly in poorly lit scenes that necessitate longer exposure time.

To tackle these challenges, two strategies have been formulated. One involves using high dynamic range (HDR) photography [4], capturing multiple low dynamic range (LDR) images at varying exposure levels and tone-mapping them to an HDR image. This technique inherently copes well with illumination changes and can capture a broader range of information than an LDR image contains. However, movements during capture can lead to inconsistencies, which prevents perfect pixel alignment in the resulting HDR image. Artifacts, which can be considered as a type of motion blur [5], ultimately lead to degraded performance in subsequent processing tasks. An alternative way entails modifying the camera's exposure [6]–[8], in response to variations in lighting conditions. By selecting an appropriate exposure allocation strategy, specifically assigning the exposure to the exposure time and the analog gain, motion blur is effectively alleviated. Although the full dynamic range of a scene is not captured, it optimizes the range to maximize information capture. Nonetheless, this method is fragile to illuminance changes and its effectiveness largely hinges on the adjustment rate and the camera's response time.

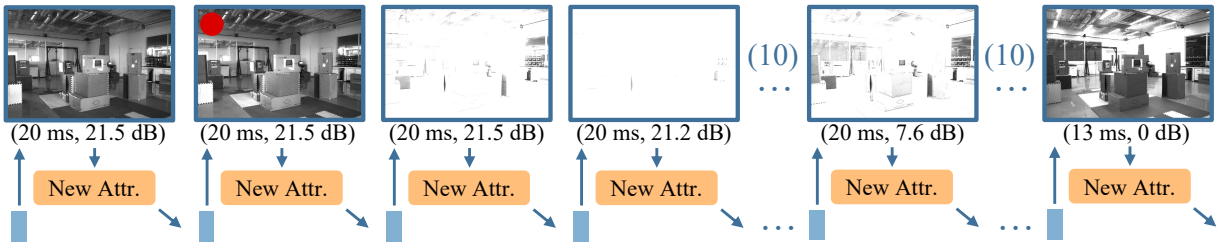
Nowadays, frame cameras can operate at high frame rates exceeding 100 Hz. However, visual SLAM tasks commonly requires an input frame rate of 20-30 Hz [9]–[11]. In this paper, we introduce VirCap, a camera capture system that requires no response time and recovers from complex lighting without delay. VirCap uses a bracketing capture patterns, engaging a higher input frame rate than traditional capture systems, and using surplus images to enhance the output quality. By integrating the previously discussed dual strategies, VirCap provides an effective solution to the three aforementioned challenges. Our contributions include:

- A novel camera exposure control framework that ad-

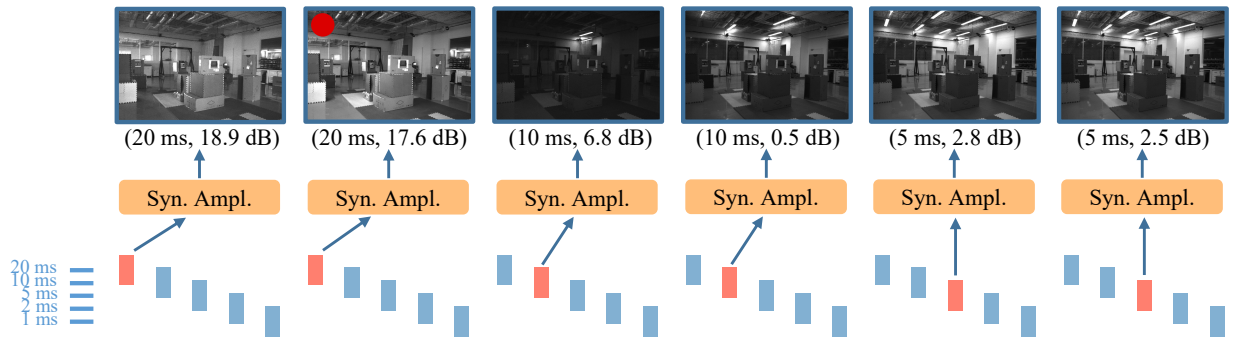
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(a) Image sequences from camera's built-in feedback control method. Owing to its *reactive* nature and the necessity to interface with the camera, it incurs a latency of 2 frames to respond to illuminance change and requires over 20 frames to recuperate from overexposure.



(b) Image sequences from our system VirCap. Unlike approaches that require alterations to camera parameters, our method utilizes synthetic amplification and can achieve a higher feedback control frequency, with 10 iterations per exposure bracket. Concurrently, as images with varying exposure time are gathered within a single bracket, the most suitable base image is selected for this synthetic amplification. Our approach is *defensible*, which eliminates the need for response time and effectively mitigates the risk of oversaturation.

Fig. 1. Image sequences in a drastic illuminance change scenario. Images with red circle indicate the first frame of illuminance change.

vances beyond the limitations of traditional ones. This framework employs photometric image synthesis techniques to achieve update and control frequencies that surpass the camera's hardware capabilities. Additionally, bracketing capture patterns are utilized to detect a broader dynamic range of information and provide quicker responses in scenes with significant lighting variations.

- an innovative exposure allocation method that effectively balances image background noise and motion blur. This method utilizes an intrinsic metric derived from a single image, avoiding the reliance on interframe information.
- Sufficient experiments demonstrating notable and robust performance improvements in scenarios involving drastic changes in illuminance and motion blur. These improvements are particularly evident in the backend application of visual SLAM.

II. Related Work

Camera exposure control methods can be categoried into gradient descent-based, global optimality-based and deep learning-based methods.

A. Gradient Descent-based Methods

Gradient descent-based methods, also known as the feedback control methods, are favored for their computational efficiency. Shim *et al.* [6] utilized γ-correction techniques to simulate images across various exposure levels. They assessed image quality through gradient magnitude and employed a fifth-order polynomial to determine the optimal gamma value. Zhang *et al.* [7] refined the image gradient metric for practical application and provided comprehensive evidence of the derivative of the image gradient concerning their metrics. Han *et al.* [12] combined a gradient-based iterative search with image simulation and integrated optical flow to gauge the camera's self-motion, facilitating gain and exposure time adjustments that are cognizant of motion blur. Gradient descent-based methods exhibit a *reactive* behavior, in which they can only respond subsequent to changes in the scene. In scenarios where illumination fluctuates sharply, oversaturation (underexposure and overexposure) is generally unavoidable. Furthermore, the frequency at which these methods update their exposure parameters is directly influenced by the camera hardware interface, which determines the duration of the oversaturation.

B. Global Optimality-based Methods

Several methods have been developed to create synthetic images with varied exposures, yielding globally optimality in a direct manner. Kim *et al.* [8], [13] crafted an image quality metric that incorporates gradient, entropy, and signal-to-noise ratio. Synthetic images are produced from a seed image based on photometric principles. By employing Bayesian optimization, they simultaneously determined the ideal exposure time and gain settings. Gamache *et al.* [14] considered to use bracket patterns to generate a benchmark for image exposure tasks. In our preceding work [15], we built upon Kim's methodology and integrated image bracketing patterns with an image synthesis technique. Our system periodically surveys a HDR spectrum of the scene to better capture

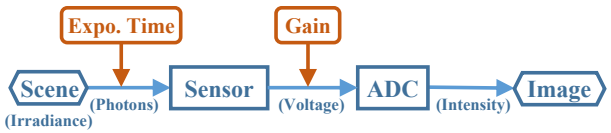


Fig. 2. A demonstration of the image generation pipeline.

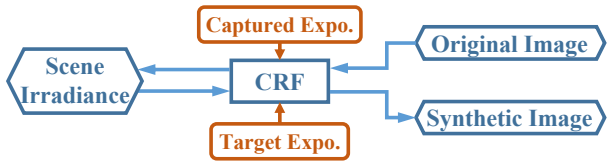


Fig. 3. Photometric image synthesis pipeline. Scene irradiance can be recovered from the exposure parameters, and virtual images with target parameters can be synthsized.

the scenario's irradiance. These global optimal methods are usually *proactive* to deliver the optimal settings within a single image capture. However, these methods have a substantial computational demand, owing to the exhaustive search for the global optimum, which also precludes the guarantee of a stable output frequency.

C. Deep Learning-based Methods

Researchers have leveraged deep learning (DL) frameworks to address the complexities associated with camera exposure control. Tomasi *et al.* [16] pioneered to integrate a neural network into a vehicle's VO system to adjust camera settings in anticipation of impending scene changes, thereby enabling intelligent and adaptive responses. They formulated a proxy criterion within a dual-camera setup to ascertain the most propitious image pairs to train their model, employing a reinforcement learning approach. Lee et al. [17] advanced a deep reinforcement learning framework that utilizes the Soft Actor-Critic (SAC) algorithm. Their framework is proficient in executing real-time processing on CPU-based devices and benefits from an expedited rate of convergence. Despite the efficacy in certain applications, DL-based methods need extensive datasets and substantial training support. Despite claims of achieving *proactive* intelligence, only a limited practical experiments and applications have verified the flexibility and scalability of these methods.

In this paper, we introduce a novel framework, VirCap, which employs image bracketing technology and introduces a virtual camera exposure control. Diverging from all previous methods, VirCap adopts a *defensive* manner (Fig. 1): It can handle drastic illuminance changes without requiring reaction time. It immediately extracts effective information from backup images and provides a reasonable output just after the change. Furthermore, VirCap decouples the exposure update process from camera acquisition cycles, thereby avoiding the need for direct interaction with the camera's control interface. This approach removes the constraints on the exposure update frequency, significantly reducing the latency typically associated with feedback control systems.

III. Photometric Image Synthesis

A. Image Generation Pipeline

We first outline the ideal imaging pipeline without noise (Fig. 2). During an exposure with a duration of t, photons keep passing through the lens and stimulating the camera sensor. The radiant flux of the photon stimulation, known as the sensor irradiance $\boldsymbol{E} = \alpha \boldsymbol{P}$, where $\boldsymbol{P}$ is the number of photons on pixels per unit time, and α is the quantum efficiency factor. The sensor irradiance is converted into a voltage signal and afterward amplified by the analog gain g. An analog-to-digital converter (ADC) transforms the real signals to the integer intensity outputs. An unknown nonlinear function $\mathcal{F}(\cdot)$ remaps the image intensity before outputting the final image $\boldsymbol{I}$ to downstream tasks. The overall process is formulated as

$$\boldsymbol{I} = \mathcal{F}\left(g \cdot \alpha \cdot t \cdot \boldsymbol{P}\right) = \mathcal{F}\left(e \cdot \boldsymbol{E}\right), \tag{1}$$

where e is defined as the exposure, used as an intermediate variable combined by exposure time t and analog gain g. It should be noted that g is always represented as decibel (dB), and it needs to convert into a multiple for calculation. In Eq. (1), we use the simple form of expression. The exposure e should be expressed completely as

$$e = t \cdot 10^{\frac{g}{20}}. \tag{2}$$

Consider introducing noise into the pipeline [18], the system can be then represented as

$$\boldsymbol{I} \sim \mathcal{F}\left(g \cdot \left(\alpha \cdot \mathcal{P}(t \cdot \boldsymbol{P}) + N_d\right) + N_r\right). \tag{3}$$

The number of photons within an image exposure follows a Poisson distribution, and the noise before and after applying the analog gain are simplified as Gaussian distribution, represented as $N_d \sim \mathcal{N}(0, \sigma_d^2)$ and $N_r \sim \mathcal{N}(0, \sigma_r^2)$, respectively.

B. Camera Photometric Calibration

Photometric calibration aims to reveal the unknown nonlinear remapping $\mathcal{F}(\cdot)$, commonly represented in its inverse logarithmic form $\mathcal{G}(\cdot) = \ln \mathcal{F}^{-1}(\cdot)$, known as *inverse response function*

$$\mathcal{G}(\boldsymbol{I}) = \ln e + \ln \boldsymbol{E}. \tag{4}$$

We refer to the method of [19], which captures a set of images with different exposure time from a static scene and outputs a lookup table for the discrete form of $\mathcal{G}(\cdot)$. We use all the pixels over $\boldsymbol{I}$ instead of several pixel samples to enhance the stability of calibration. Due to the increase of sample volumn, we replace the original SVD-based solution with Ceres solver [20]. Outlier pixels, containing a certain amount of boundary values (0 and 255) are removed to improve the fitting accuracy. After calibrating the discrete lookup table, the function $\mathcal{G}(\cdot)$ and its inverse $\mathcal{G}^{-1}(\cdot)$ are parameterized by tenth polynomial functions.

C. Photometric Image Synthesis

As shown in Fig. 3, the synthesis process from an original image $\boldsymbol{I}_0$ of exposure e_0 to a target image $\boldsymbol{I}_1$ of exposure e_1 can be defined as

$$\begin{aligned}\boldsymbol{I}_1 &= \mathcal{S}(\boldsymbol{I}_0, e_0, e_1) \\ &= \max\left(0, \min\left(255, \mathcal{G}^{-1}(\mathcal{G}(\boldsymbol{I}_0) - \ln e_0 + \ln e_1)\right)\right).\end{aligned} \tag{5}$$

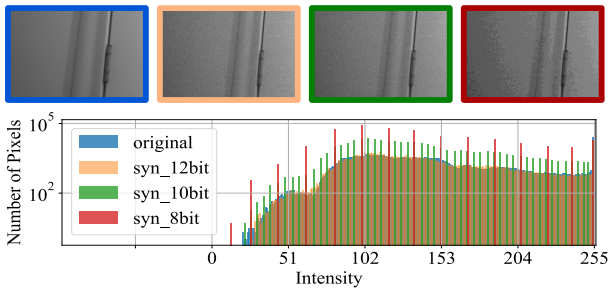
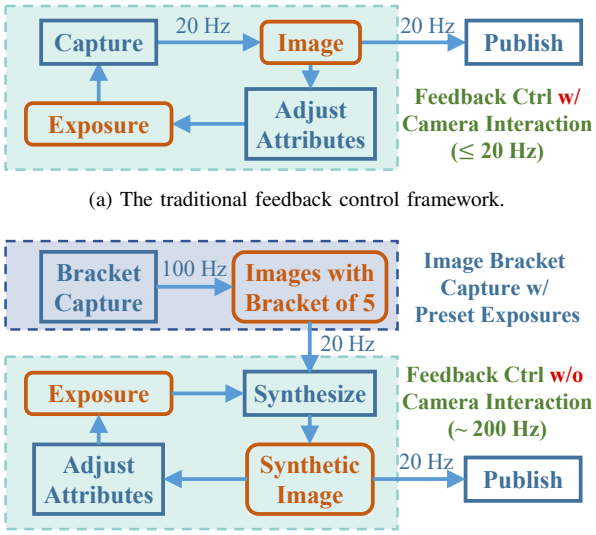


Fig. 4. Image synthesis using ADCs with varying bit depths. The original image was captured with an exposure time of **40** ms, whereas the base image for the synthesis process was captured at **1** ms. The original image encompasses **234** distinct intensity values. In contrast, the synthesized images with different bit depths, namely **12**-bit, **10**-bit, and **8**-bit ADCs, offer **228**, **54**, and **15** intensity values, respectively. It is observed that the quantization error becomes increasingly pronounced with the utilization of lower bit-depth ADCs.



(a) The traditional feedback control framework.

(b) Our proposed framework with virtual camera control. Camera exposure update is stripped from feeback control, while our loop can execute at a very high frequency.

Fig. 5. The comparison between traditional camera control and our proposed virtual camera control frameworks.

The synthesis process is similar to the amplification with analog gain. The main difference is that the photometric synthesis process amplifies the signals after the ADC and brings in more source of noise. In our previous work [15], we found that the images synthesized from high to low exposures are not reliable, since the information has been already lost and the overexposed areas cannot be recovered. On the other direction, the synthetic process is reasonable. The quantization errors from the ADC, caused by the precision loss from real values to integers, are revealed only when the amplification ratio increases. One way to ralleviate the quantization error is to use ADCs with higher bits (Fig. 4). Photography and industrial cameras usually come with high-bit ADCs, and have already been used in some image processing tasks [21], [22]. We use industrial cameras with 12-bit ADC and set the maximum synthesis amplification ratio as 100, equivalent to an analog gain of 40 dB.

IV. Virtual Camera Control

The traditional framework for updating camera exposure is depicted in Fig. 5a. A significant limitation of this framework is that the control frequency is capped by its acquisition frequency, as the system relies on the collection of actual images to obtain feedback. Additionally, modifying camera exposure parameters needs interaction with the control interface, which typically incurs a latency of 1 to 2 frames due to the interactive communication. Additionally, the system is *reactive*, that it cannot predict lighting shifts in the future, where images may become instantly overexposed or underexposed, and it can only gradually recovers in subsequent frames.

We introduce a pioneering framework, VirCap (Fig. 5b), that obviates the need for direct interaction with the camera for modifying exposure parameters. VirCap employs a bracketing pattern to capture images at varying exposures, utilizing a synthesis technique to generate images with the desired exposure for the same scene. This approach allows the expected exposure to be updated independently from the camera hardware, leveraging the synthesis technique to unbind and enhance the update frame rate. Furthermore, the challenge of exposure allocation—balancing exposure time and analog gain—is addressed by selecting an optimal image with a specific exposure time from the bracket. Subsequently, the equivalent analog gain is calculated based on this selection. The output image is synthesized from the chosen base image and is released for each bracket, ensuring that the most appropriate exposure settings are consistently applied and maintained.

A. Camera Exposure Update

Like other camera exposure control methods, VirCap also requires an exposure update module to determine the optimal exposure in real time. Instead of designing a new update module from scratch, we opt to adopt the exposure update module developed by Shim [6]: This method obtains images under different gamma values through γ-correction technique, and calculates the gradient magnitude as an image quality metric to find the numerical relationship between this metric and gamma values. It uses the optimal gamma value as a reference to calculate the corresponding updated exposure. It should be noted that VirCap is flexible and can incorporate most exposure update methods based on feedback control.

Here, we introduce the complete virtual update process implemented in VirCap:

1) Initialization: VirCap begins by capturing a bracket of images. It then calculates the image quality metric as defined in the update module—specifically in our configuration, the image gradient from the method of Shim. The relationship between this quality metric and the exposures is regressed to determine the optimal exposure, which is selected as the initial target exposure e_0.

2) Update Cycle: When entering the update cycle with a bracket of images and a target exposure of e_i, the allocation module, which will be discussed in Sect. IV-B, firstly selects a reference image from the bracket with an exposure time t_i. The reference image is then amplified to the target exposure

e_i, using the synthesis technology described in Sect. III-C. Afterwards, the synthetic image is processed using γ-correction to find the most suitable γ_i for update. Finally, The new target exposure e_{i+1} is updated using

$$e_{i+1} = e_i \cdot (1 + k \cdot (1 - \gamma_i)), \tag{6}$$

where k is to balance the convergence speed and stability. The implementation details of exposure update can be found in [6].

Since VirCap derives its control feedback from synthesized images rather than from actual image collection, it does not need to wait for camera's hardware to update the exposure. The upper bound of the control frequency is determined solely by the synthesis process and exposure update method. We conduct 10 iterations for each bracketing update, allowing the update frequency to reach up to 200 Hz. Furthermore, to conserve resources, we integrate an early stopping mechanism that halts the update if no change in exposure is detected within a bracket.

B. Exposure Allocation Strategy

Upon determining the optimal exposure e through the preceding update process, it is necessary to decompose this value into two independent control parameters: **exposure time** t and **analog gain** g. Traditional camera systems often employ methods that prioritize either gain or exposure time. In our previous research [15], we developed a stepped allocation algorithm to regulate the rate of increase for each exposure, thereby preventing any single exposure from escalating excessively. Kim [8] proposed a metric that considers image gradient, image entropy, and background noise, and employed Bayesian Optimization (BO) for the allocation process. Furthermore, Han *et al.* [12] harnessed optical flow to gauge the magnitude of the camera's intrinsic motion, thereby establishing a linkage with the allowable maximum exposure time.

Our method departs from traditional single-image capture, as we capture a bracket of potential image candidates. Our task is reformulated: Initially, we define a specific metric to evaluate image quality. Afterwards, we perform calculations using this metric across all images in the captured bracket, and select the image with highest score as the base image (with an exposure time t). Finally, we calculate the corresponding target analog gain g via

$$g = 20 \times \log_{10}(\frac{e}{t}) \tag{7}$$

According to the nature of moving robots and complex illumination scenarios, we develop an image quality metric that simultaneously considers for motion blur and background noise. Unlike methods that rely on interframe information, such as optical flow, to assess the extent of the camera's motion, we use intrinsic information from individual images.

For quantifying motion blur, we directly employ the non-reference blur metric developed by Crete et al. [23], denoted as $\mathcal{M}_b$. This metric scales between 0 and 1, assessing the degree of image blur by quantifying the loss of high-frequency information due to low-pass filtering. Nevertheless, background noise has the potential to artificially increase high-frequency information, which could be misinterpreted as sharpness or texture details. We precede metric evaluation with a boxfilter denoising process to counteract this effect. Although motion blur might be slightly overestimated in images with minimal blur, we deem this impact to be within tolerable ranges for our application. The noise level, denoted as $\mathcal{M}_n$, is contingent upon the camera's imaging principles. It correlates with both exposure time and analog gain, and remains independent of the image content itself. Referencing the image generation pipeline detailed in Section III-A, we adopt the simplifications presented in [18], [24] to represent the Poisson distribution $\mathcal{P}(t\boldsymbol{P})$ using an equivalent Gaussian distribution $\mathcal{N}(t\boldsymbol{P}, t\boldsymbol{P})$. The random variable in Eq. (3) can be transformed as

$$\begin{aligned} \boldsymbol{I} &\sim \mathcal{F}\left(g \cdot (\alpha \cdot \mathcal{P}(t \cdot \boldsymbol{P}) + N_d) + N_r\right), \\ \exp \mathcal{G}(\boldsymbol{I}) &\sim g \cdot \alpha \cdot \mathcal{P}(t \cdot \boldsymbol{P}) + g \cdot N_d + N_r, \\ \exp \mathcal{G}(\boldsymbol{I}) &\sim g \cdot \alpha \cdot \mathcal{N}(t\boldsymbol{P}, t\boldsymbol{P}) + g \cdot N_d + N_r, \\ \operatorname{Var}\left[\exp \mathcal{G}(\boldsymbol{I})\right] &= g^2\alpha^2 t\boldsymbol{P} + g^2\sigma_d^2 + \sigma_r^2. \end{aligned} \tag{8}$$

We determined that the exposure time has a minimal impact on background noise, and the first term in Eq. (8) plays an insignificant role in noise generation, leading us to disregard it for simplification purposes. The noise level is modeled as a quadratic function in relation to the analog gain g,

$$\mathcal{M}_n = \operatorname{Var}\left[\exp \mathcal{G}(\boldsymbol{I})\right] \approx g^2\sigma_d^2 + \sigma_r^2, \tag{9}$$

and the noise parameters σ_d and σ_r need to be estimated. To regress the noise parameters, we selected a static scene and captured a series of images with different exposure combinations. The comprehensive image metric, considering both motion blur and background noise, is formulated as a combined measure of these two metrics

$$\mathcal{M} = \mathcal{M}_b + \beta\mathcal{M}_n, \tag{10}$$

where β is a weighting factor that balances the significance of each component.

C. Prevention against Drastic Illuminance Change

Since our system continues to utilize a feedback control-based method, it inherently needs multiple iterations to update the camera exposure when drastical change happens. This brings computational burden and affects the real-time performance of update process.

Considering that the image bracket has HDR scene information, we incorporate a restart mechanism: When the average pixel intensity difference between two adjacent frames using the same exposure is greater than a predefined threshold $\mathcal{T}_{\boldsymbol{I}}$, the system determines that severe lighting has occurred. Our system firstly processes and releases the images captured before the shift. After capturing a new bracket, our system falls back to the initialization process in Sect. IV-A.

V. Experimental Evaluations

A. Experimental Setup

Our experimental device is shown in Fig. 6. Four industrial cameras (FLIR BFS-PGE-04S2M) with a image resolution of 720×540, deployed our capture method and other baseline methods, respectively. The capture methods are implemented

TABLE I
PERFORMANCE COMPARISON WITH ORB-SLAM3 USING POSE APE (RMSE). THE MARKER '-' REPRESENTS THE FAILURE CASE.

Seq.	*Bright*				*Dark*				*Switch*			
	01	02	03	04	01	02	03	04	01	02	03	04
Kim	0.14	0.23	0.20 (2)	0.17 (2)	0.12	0.28	0.09 (2)	-	0.09 (2)	-	0.09	1.15 (7)
Shim-G	**0.08**	**0.10**	**0.04**	0.09 (2)	0.10	**0.06**	0.07 (1)	**0.10**	0.09 (3)	-	0.06 (2)	-
Shim-E	0.12	0.35	**0.04**	**0.08** (2)	0.24 (1)	0.12	0.09	0.12 (1)	0.16 (4)	-	**0.05** (2)	-
Ours	0.11	0.12	0.07	0.11 (2)	**0.08**	0.07	**0.05**	0.13 (1)	**0.08** (2)	**0.19** (1)	0.06 (1)	**0.14** (2)
Platform	Hand.	Hand.	Veh.	Quad.	Hand.	Hand.	Veh.	Quad.	Hand.	Hand.	Veh.	Quad.
Dist. (m)	55.70	64.10	58.83	54.25	47.77	61.07	58.31	46.49	88.96	66.64	90.26	93.57
Dura. (s)	110.98	81.00	58.80	87.97	91.37	123.75	68.00	110.99	146.69	79.56	95.23	99.99
Velo. (m/s)	0.50	0.79	1.00	0.62	0.52	0.49	0.86	0.42	0.61	0.84	0.95	0.94



Fig. 6. Our handheld device, and the robot platforms.

TABLE II
PERFORMANCE COMPARISON WITH DPVO USING ROTATION RPE (RMSE / MAX).

Seq.	Kim	Shim-G	Shim-E	Ours
B01	0.067 / 0.585	**0.062** / **0.328**	0.064 / 0.594	0.064 / 0.390
B02	0.107 / 1.238	0.112 / 1.570	0.104 / 1.433	**0.095** / **0.656**
B03	0.101 / 0.642	**0.096** / **0.408**	0.110 / 1.014	0.112 / 1.010
B04	0.087 / 1.786	0.087 / 1.801	**0.086** / **1.771**	0.087 / 1.804
D01	0.072 / 0.681	**0.062** / 0.263	0.063 / 0.266	0.064 / **0.201**
D02	0.065 / 0.388	**0.050** / 0.267	**0.050** / 0.272	0.051 / **0.261**
D03	0.105 / 0.928	0.140 / 2.643	0.095 / 1.462	**0.092** / **0.769**
D04	0.069 / 0.787	**0.054** / **0.163**	0.054 / 0.167	0.065 / 1.100
S01	0.068 / 0.839	**0.061** / **0.779**	**0.061** / 0.787	0.062 / 0.847
S02	0.122 / 1.488	0.132 / 1.612	0.142 / 1.859	**0.116** / **1.148**
S03	**0.131** / **1.704**	0.184 / 2.577	0.149 / 2.458	0.147 / 1.895
S04	0.161 / 1.631	0.139 / 1.258	0.139 / **1.199**	**0.137** / 1.525

on a mini PC with an Intel NUC11TNKi7 CPU and 16 GB RAM. We build the device as a handheld platform, and also equip it on a mobile car robot and a quadrupedal robot for different experimental tasks. The ground truth is given by a OptiTrack motion capture system, time synchronized to our device via local area network.

Several baseline methods are implemented for comparison:

- The **Built-in** method, embedded in the camera hardware, follows a feedback control rule to make the average image intensity tend to a preset value.
- The method of **Shim** [6] uses the γ-correction techniques to simulate images with different gamma levels. Since they only considered the exposure time instead of both the exposure time and the analog gain, we implement an exposure allocation module with exposure time priority (**Shim-E**), gain priority (**Shim-G**), and optical flow-based strategy (**Shim-OF**) from [12], respectively.
- The method of **Kim** [8] is a globally optimal baseline method. They designed a image quality metric combined with image gradient, image local entropy, and image noise. Their attribute allocation is introduced by their Bayesian Optimization method.

Due to the limited number of available camera devices, we are unable to run all baseline methods concurrently in each experiment. Therefore, for each experimental task, we choose three methods to benchmark against our own.

Within the configuration of our VirCap, there are 5 levels of exposure time in a bracket: 1 ms, 2 ms, 5 ms, 10 ms, and 20 ms. The camera in our experiment can run at a frame rate of 100 Hz, and the system output can reach to 20 Hz. It is important to note that the set of bracketing exposures can be adjusted based on user requirements. If perceiving a broader dynamic range is required, the interval between two adjacent exposures can be increased, but the quality of the synthetic images may be reduced. We set the parameter to limit each bracket's updates to a maximum of 10 times in order to control overhead. For initialization and cases involving drastic illumination changes, we increase the update limit to 30 to allow for a more thorough update.

B. Evaluation: Visual SLAM

An efficacious visual SLAM system hinges on the prerequisite that the captured images possess abundant feature information and sustain a robust correlation between successive frames. Particularly in low-light conditions, an extended exposure duration coupled with elevated analog gain becomes imperative, which concurrently escalates the level of background noise. Additionally, given the inherent nature of mobile robots, motion blur emerges as a significant determinant impeding feature extraction and thus warrants careful consideration.

We conduct the acquisition of a suite of image sequences under varying lumination conditions utilizing different robotic platforms. The *Bright* sequences are captured with the lights maintained within the experimental site, achieving an illuminance level of approximately 600 lx. Conversely, the *Dark* sequences are recorded with the lights extinguished, plunging the scene into profound darkness (less than 20 lx). Notably, the dark scenarios exhibit an expanded dynamic range due

to intense light, infiltrating the experimental site from adjacent rooms through the surrounding windows. The *Switch* sequences are characterized by the intermittent toggling of lights in the experiment. The employed capturing methods need resilience against the degradation of precision or the potential breakdown of inter-frame linkage attributable to abrupt changes in illuminance.

Two visual SLAM systems are deployed on these sequences obtained through various capturing methods. ORB-SLAM3 [25] is a prevalent visual SLAM framework that operates by extracting ORB [26] features and tracking them across frames. DPVO [27], on the other hand, is an innovative visual odometry system underpinned by a Recurrent Neural Network (RNN). Diverging from the traditional approach of descriptor extraction and tracking, DPVO harnesses deep features to deduce the optical flow from selected image patches. Given that ORB-SLAM3 incorporates loop closure detection and relocation modules to mitigate tracking losses, the RMSE of the absolute pose error (APE) is employed as a performance measure, and the number of system restarts throughout the sequences is also recorded. In contrast, DPVO lacks a loop closure detection feature. While it is typically robust against tracking loss, it can occasionally yield drastically incorrect estimates. Moreover, our findings indicate that DPVO is generally susceptible to scale drift during its operation. Therefore, the relative pose error (RPE) is taken as a metric, with only the RMSE and the maximal drift on rotation being evaluated. The assessments are conducted using the EVO [28] tool suite, with the outcomes for ORB-SLAM3 and DPVO presented in Table I and Table II, respectively.

In well-illuminated scenarios (*Bright* sequences), all four capturing methods exhibit comparable performance across both visual SLAM systems. It is noteworthy that the **Shim-G** method consistently outperforms the other three in such brightly lit scenarios. This is because the background noise is minimal in well-lit conditions, despite the application of extremely high analog gain, Alternatively, motion blur remains the primary element that impacts the efficacy of front-end feature tracking, a challenge that the greed gain prior strategy can substantially mitigate. **Ours** slightly underperforms compared to **Shim-G**. The difference is minimal and primarily noticeable at the centimeter level in RMSE measurements. This discrepancy, although an order of magnitude in the context of SLAM problems, is not readily apparent.

Within the context of *Dark* sequences, the methods **Shim-G**, **Shim-E**, and **Ours** demonstrate indistinguishable performance when applied to both visual SLAM systems, achieving commendable outcomes. The approach proposed by **Kim** requires the insertion of seed frames during the acquisition process to facilitate exposure updates. The seed image mechanism is frequently invoked due to the HDR nature of *Dark* sequences and the substantial fluctuations in image brightness. Consequently, their frame rate is halved to approximately 10 Hz. DPVO boasts a robust front-end tracking module, and the performance of **Kim** in conjunction with DPVO is comparable to that of the other methods. However, in the case of ORB-SLAM3, which relies on traditional feature extraction and matching that proves to be fragile under complex lighting conditions, there is a noticeable decline in performance. This degradation can escalate to complete failure in scenarios involving extensive movements and large rotation (*Dark-04*).

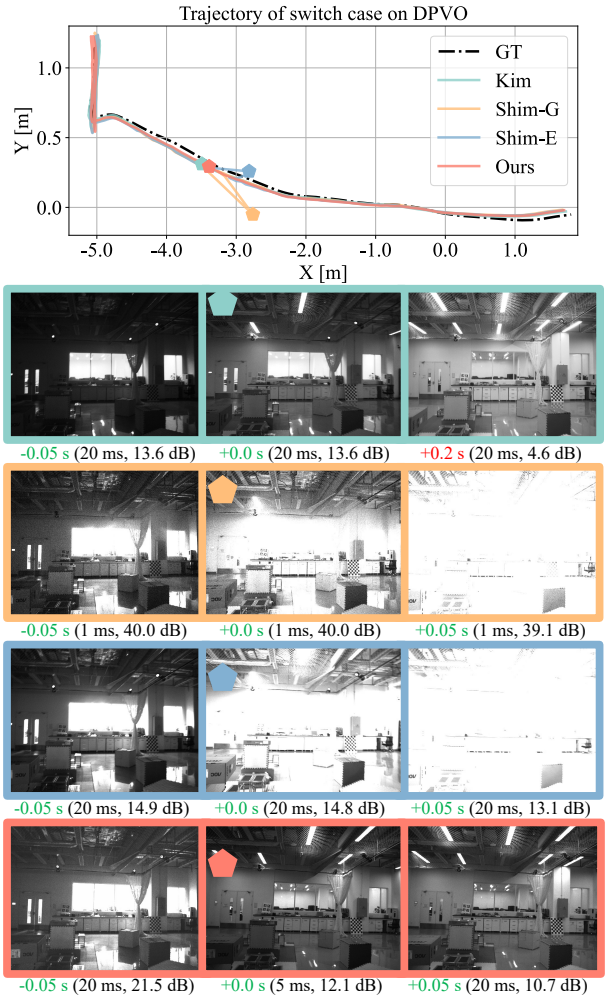


Fig. 7. Trajectory on DPVO of a light switch case. The image at the moment of sudden illuminance change and its frames before and after are also given. The exposure parameters (exposure time and analog gain) with the timestamp offset referred to the light switch frame are given.

In *Switch* sequences, where abrupt transitions in illumination are prevalent, **Kim**'s method demonstrates notable adeptness in accommodating rapid changes in lighting. This effectiveness is largely due to its one-step optimization strategy. Apart from isolated failure cases of *Bright-02* and *Bright-04* on ORB-SLAM3, resulting from a decrease in frame rate and weak inter-frame associations due to excessive illuminance variations, their approach yields favorable outcomes. Conversely, methods based on **Shim** are constrained by the principles of feedback control, necessitating a certain duration to recover from overexposure when the lighting is intensified and from underexposure when it is diminished. This feedback control approach is ineffective under such lighting conditions on ORB-SLAM3, leading to complete failure. Given that DPVO incorporates a robust front-end optical flow association, these methods manage to function, albeit with reduced efficiency, owing to the oversaturation effect. The method of

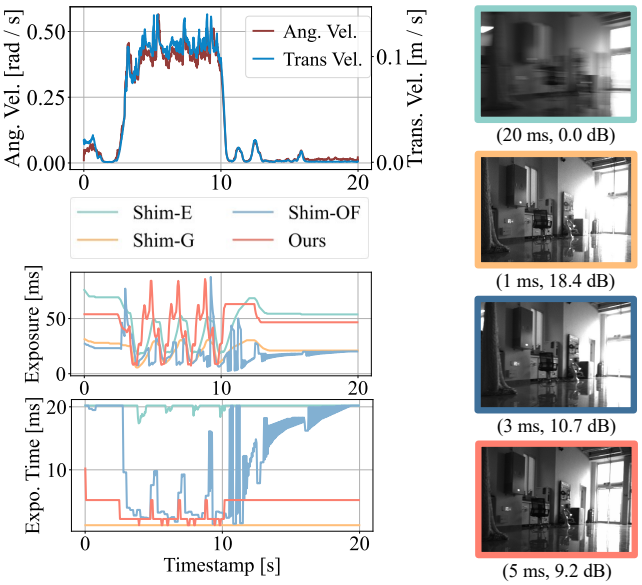


Fig. 8. Statistical information and images of the robot vehicle rotation sequence. Our method responds faster from the overexposure and has more timely control of exposure compared to the three other methods.

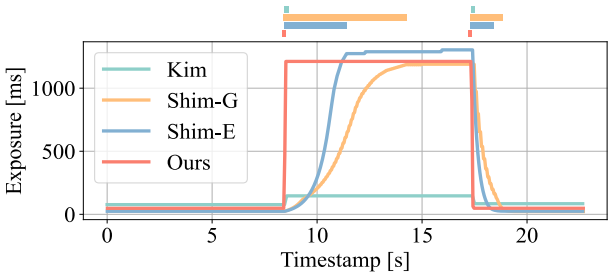


Fig. 9. A sudden illuminance change cases. The bar over the diagram shows the start time and duration of the exposure control. **Ours** recovers without any delay to provide an informative image, while other methods require **1** frame to perceive the illuminance change. **Kim** gaps an extra frame for its seed mechanism.

Ours also employs a feedback control mechanism; however, our method distinguishes itself by decoupling the feedback control process from the camera hardware, thereby facilitating more rapid exposure adjustments with minimal oversaturation. Consequently, our method maintains commendable stability and performance across all *Switch* sequences.

A demonstrative example of the DPVO under a lighting turing on scenario is depicted in Fig. 7. Methods of **Shim** are unable to adapt to the abrupt onset of intense illumination, culminating in significant overexposure in subsequent images and immediate discontinuities in the trajectory estimation. Conversely, **Kim**'s methodology requires only a single frame to respond, rapidly converging to the globally optimal settings for the altered light conditions. However, their response is 1 frame slower, and still under the risk of overexposure. **Ours** similarly exhibits prompt responsiveness. By selecting an appropriate base image from an array of images captured at varying exposure levels, our approach obviates the need for a multi-frame delay to gauge the scene's illuminance response. Instead, it immediately identifies a suitable new base image from the available candidates, effectively addressing the challenges posed by sudden lighting shifts.

C. Evaluation: Exposure Allocation Strategy

To evaluate the effects of different exposure allocation strategies, we equip the mobile car robot platform with the necessary hardware and program the robot to perform a rotation on the spot, completing a full turn approximately every 2 seconds. This setup allows us to generate significant yaw axis rotation. The ground truth of the angular and translational velocity is captured and calculated using the OptiTrack motion capture system, which serves as a reference for assessing image blurriness. Alongside our method, we implement three baseline exposure control methods based on Shim's approach, each employing a distinct allocation strategy: **Shim-E**, **Shim-G**, and **Shim-OF**.

The statistical information of the exposure e and exposure time t is depicted in Fig. 8. Methods utilizing optical flow can effectively respond to the camera's movement and allocate parameters according to the degree of motion. However, this system meets severe oscillations. This is because accurately capturing optical flow becomes challenging in environments with complex lighting and dynamic motion, subsequently reducing the stability of the allocation algorithm. Enhancements can be made by finely tuning system parameters and incorporating additional safeguards to mitigate these effects. Our system circumvents this issue as it relies solely on single-frame images to assess motion, without utilizing inter-frame information. This ensures the stability, avoiding the complications that arise with methods dependent on optical flow, especially in challenging lighting and motion conditions.

D. Evaluation: Camera Response Speed

We also evaluate the ability of different methods to cope with transient illumination changes. We choose the same baseline methods as in the previous Sect. V-B. We fix the device in a static scene and observe the response speed of each control methods by change the illuminance (turning on and off the lights) in the scene. The scene presents extreme lighting conditions. All lights in the environment switch on and off almost simultaneously. When the lights are on, the illuminance reaches 600 lx, and when they are turned off, it drops to around 0 lx.

The results are displayed in Fig. 9, with additional visual demonstrations available in Fig. 1 and Fig. 7. We observe that methods using traditional feedback control frameworks require a certain time to stabilize after a sudden disturbance. Specifically, **Shim-G** and **Shim-E** take approximately 6.5 seconds and 3.0 seconds, respectively, to recover from underexposure, and 2.0 seconds and 1.5 seconds to adjust after overexposure. This difference in recovery times is attributable to the time required for the camera hardware to modify these parameters, and the difference within each methods to underexposure and respond overexposure is due to Shim's exposure settings. In contrast, both **Kim** and **Ours** exhibit the ability to quickly adapt to sudden lighting changes. **Kim**'s method updates within 0.1 seconds, accounting for one frame to perceive

TABLE III
AVERAGE TIME CONSUMPTION OF PROCEDURES

Procedure	Exp. Update	Img. Sel.	Full Synth.	Total
Duration (ms)	10.864	8.675	1.674	21.213

TABLE IV
COMPARISON OF

Method	Build-in	Kim	Shim	Ours
Output Freq. (Hz)	20.24	16.45	20.22	19.63
Exp. Update Freq. (Hz)	20.24	1.26	10.15	153.27
Avg. Time per Frame (ms)	–	14.46	1.38	21.21

the change and another frame to adjust using a seed image. However, their system may not provide a correct exposure estimation due to the significant difference in illumination before and after the light is turned on. Using a seed image with a fixed exposure in such a scenario does not guarantee a correct synthesis process within their framework.

Ours demonstrates rapid response and update speed. When the sudden lighting change occurs, information with different exposures is backed up. Our system can immediately process this backup. This ability significantly enhances the responsiveness of our system and reduces latency in adapting to new lighting conditions.

E. Evaluation: Time Consumption

We also analyze the time consumption of our method, as presented in Table III. To realize real-time implementation, we implemented several system optimizations. First, we reduced the image resolution to one-quarter of its original size, performing full-size synthesis only for the final output. Second, to accelerate the synthesis process, we employed Single Instruction Multiple Data (SIMD) techniques, achieving a synthesis speed of approximately 0.5 ms per image.

We also compare the output frame rates and time consumption of various methods. The **Build-in** method, embedded in the camera hardware, and **Shim**, both provide stable output frame rates. However, since **Shim** relies on a software interface for control, its actual frame rate drops to nearly half. Unlike feedback control methods, **Kim**'s method updates only once per scenario. Its parameters are seldom updated, leading to a low update frequency and minimal computational overhead. However, **Kim** requires a seed image during capturing, which inevitably lowers the output frequency. **Ours** method exhibits the longest time consumption but ensures real-time system operation. By utilizing an image bracket with a virtual update framework, our approach eliminates the need for hardware control, allowing for the highest exposure update frequency free from hardware limitations. Theoretically, the update frequency should be 10 times the output frequency, though it is lower in practice due to our early stopping mechanism.

VI. DISCUSSIONS

For visual SLAM tasks, numerous recent approaches [29]–[31] tackle the problem of interframe tracking loss by incorporating advanced relocation modules. We offer another perspective by prioritizing image acquisition, a fundamental module that precedes the visual SLAM system. By enhancing the stability and reliability of the captured image sequence, tracking failures are effectively reduced and this significantly increases the performance of a visual SLAM system.

Our camera exposure control method boasts exceptional capabilities for mobile robots operating in complex lighting scenarios. From an image quality perspective, while a gain prior camera control method is generally sufficient for moderate cases, where motion blur is the primary concern and background noise is less critical due to advancements in modern cameras. Our allocation method takes both factors into account. In practical applications, the parameter β in Eq. (10) can be adjusted to strike an optimal balance between these two factors, tailored to the specific needs of the task. Concerning response speed, our framework holds inherent advantages, enabling rapid responses in environments characterized by high ego-motion, extensive dynamic range, and significant illumination changes.

Our system also presents several limitations. Firstly, it requires accurate calibration of the camera's photometric parameters, which can be practically challenging. Additionally, despite using a high-bit ADC, quantization errors are inevitable. When employing high amplification synthesis (greater than 38 dB, equivalent to an 80x magnification), the resulting images exhibit more noise compared to standard imaging. Moreover, our system updates exposure and then publishes synthetic images, rather than publishing original images first and updating exposure for the next image. This mode introduces a certain output delay, approximately 20 ms in our experimental setings, which needs to be considered for robotic tasks that require high real-time responsiveness.

VII. CONCLUSIONS

In this paper, we have introduced VirCap, a novel camera exposure control system that utilizes image bracketing capture and photometric synthesis techniques. By decoupling camera exposure control from the camera control loop, our system enables more frequent exposure updates. Thanks to the bracketing mechanism, VirCap can select the most suitable base image from various exposure levels for generation, thus avoiding the oversaturation issues typically caused by drastic illumination changes in traditional feedback control systems. We have developed an exposure allocation strategy that relies on a single image frame to effectively balance motion blur and background noise. Extensive experiments have demonstrated that VirCap exhibits excellent performance and possesses distinctive characteristics, making it highly effective for various types of robots operating in complex lighting environments.

In our future research, we plan to integrate our capture system with a specific visual backend tack. We aim to leverage the bracketing sequence, which encapsulates the scene's HDR information, directly in the visual SLAM system.

ACKNOWLEDGEMENT

The authors would like to thank the Robotics and Autonomous Systems (ROAS) of the Hong Kong University of

Science and Technology (GZ), for the generous provision of experimental site.

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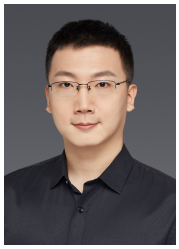
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